



Case study–topic 1 and 3

Team 3



1. Runtime Environment Mismatch

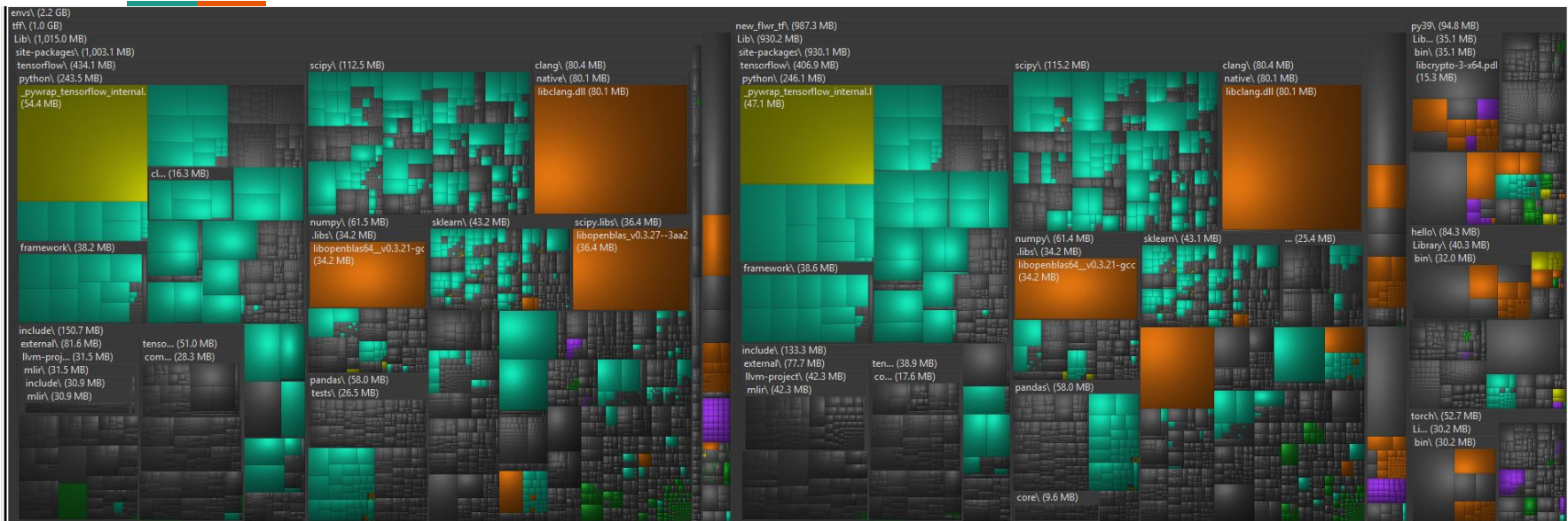
Predictions from the staging server differ significantly from the development notebook.

The staging environment uses Python 3.10, while the model was trained using Python 3.8, and several ML library versions differ.



Solution – Use Virtual environment

1. Match Python versions: Run the staging server on the same Python version (3.8) or retrain the model under 3.10 for consistency.
2. Align library versions: Freeze exact versions from your training environment (`pip freeze > requirements.txt`) and install the same versions on staging.
3. Use environment reproducibility: Create a virtual environment or use Conda/Docker so both training and staging run identical dependency stacks.
4. Re-export the model: If upgrading libraries is unavoidable, retrain or at least re-serialize the model under the new versions.
5. Add version checks: Log Python and package versions at inference time so mismatches get caught immediately.



	% of parent	Size	Allocated	Items	Files	Folders	Modified	Attributes
envs	18.6 %	2.4 GB	2.2 GB	99,559	87,755	11,804	08-11-2025 9.55.09 AM	
tff	46.4 %	1.0 GB	1.0 GB	41,045	35,830	5,215	28-05-2025 9.59.32 PM	
Lib	96.1 %	977.6 MB	1,015.0 MB	38,333	33,222	5,111	28-05-2025 9.59.32 PM	
Library	2.9 %	39.4 MB	30.7 MB	224	217	7	28-05-2025 7.01.43 PM	
Scripts	0.3 %	3.8 MB	3.7 MB	43	43	0	28-05-2025 9.59.32 PM	
tcl	0.3 %	6.8 MB	3.6 MB	1,312	1,265	47	28-05-2025 7.01.45 PM	
conda-meta	0.2 %	2.2 MB	2.2 MB	11	11	0	28-05-2025 7.01.49 PM	
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DLLs	0.0 %	6.7 MB	132.0 KB	33	33	0	28-05-2025 7.01.43 PM	
share	0.0 %	504.1 KB	8.0 KB	632	607	25	28-05-2025 8.48.19 PM	
Tools	0.0 %	951.6 KB	0	208	199	9	28-05-2025 7.01.43 PM	
libs	0.0 %	524.8 KB	0	3	3	0	28-05-2025 7.01.44 PM	
include	0.0 %	675.0 KB	0	160	158	2	28-05-2025 7.01.44 PM	
etc	0.0 %	863 Bytes	0	10	7	3	28-05-2025 7.01.49 PM	
new_flwr_tf	43.4 %	1,006.8 MB	987.3 MB	38,247	33,051	5,196	29-05-2025 2.32.53 PM	
py39	4.2 %	126.4 MB	94.8 MB	6,769	6,316	453	28-05-2025 8.18.47 PM	
hello	3.7 %	128.4 MB	84.3 MB	6,681	6,220	461	30-05-2025 3.30.20 PM	
torch	2.3 %	126.4 MB	52.7 MB	6,769	6,316	453	02-06-2025 3.55.56 PM	
tf_gpu	0.0 %	152 Bytes	0	5	3	2	18-04-2025 8.59.39 AM	
semmy	0.0 %	152 Bytes	0	5	3	2	24-03-2025 1.01.44 PM	
semantra_env	0.0 %	154 Bytes	0	5	3	2	14-03-2025 8.14.43 AM	
semantra	0.0 %	154 Bytes	0	5	3	2	23-03-2025 11.50.09 PM	
RAG	0.0 %	149 Bytes	0	5	3	2	11-06-2025 7.09.50 PM	
qlq_env	0.0 %	149 Bytes	0	5	3	2	02-06-2025 9.56.03 AM	



3. Model Risk Evaluation Concerns:

The auditor notices the model performs poorly for claims involving “rare surgical procedures” due to insufficient representation in the training data.



Solution – 2 stage Classifier model

1. For all the data that is currently present, split it into healthy and diseased, train a classifier on that. If the classifier diagnoses diseased send it to another classifier
2. The second classifier is exclusively trained on all the different cases of the diseased issue, eg: for heart disease→ arrhythmia, x,y,z only. It has no case of a healthy heart.
3. Now, the ratio of all rare diseases is uniformly distributed, the accuracy of prediction of a rare disease increases significantly.
4. This two-stage pipeline reduces noise at every step. The first classifier only focuses on distinguishing “normal vs abnormal,” so it becomes extremely good at identifying deviations, even subtle ones, without being distracted by the complexity of multiple disease types.

Heart Condntions

Heart valve disease

4.7%

Cardiomyopathy

1.6%

Atrial Fibrillation

8.5%

Angina

7.0%

Coronary artery dise...

1.6%

Congenital heart dis...

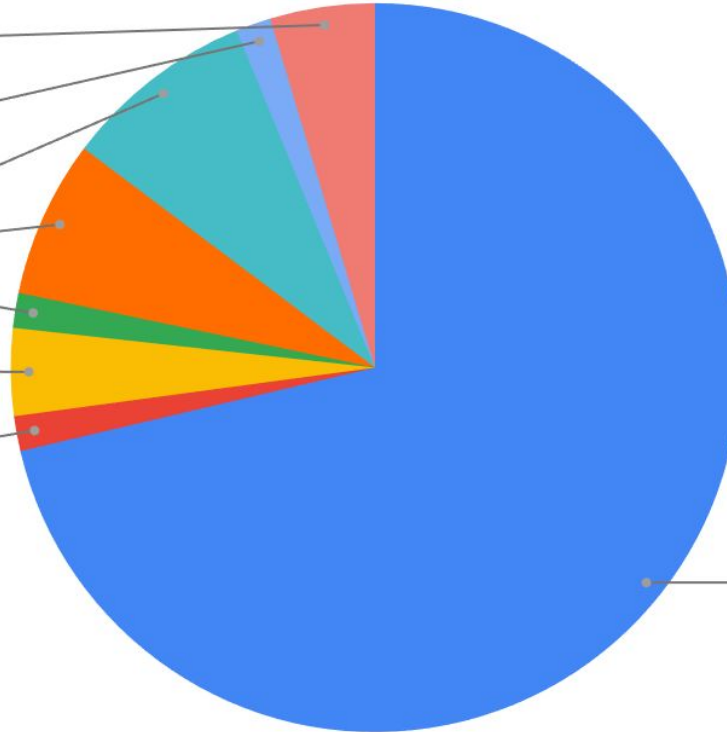
3.9%

Arrhythmia

1.6%

Normal

71.3%



Heart valve disease
21.4%

Angina
32.1%

Cardiomyopathy
7.1%

Atrial Fibrillation
39.3%

